Algorithm and Data Structures: Problem 14t Solution

Question 15. *We define the* iterated logarithm function

$$\lg^*(n) := \min\{i \in \mathbb{N} \mid \lg^{(i)}(n) \leq 1\},$$

where $\lg^{(1)}(x) := \lg x$ *and* $\lg^{(k+1)}(x) := \lg(\lg^{(k)}(x))$.

This function grows extremely slowly. Order the following functions by asymptotic growth:

$$\begin{array}{cccccc}
\lg\lg^*(n) & 2^{\lg^*(n)} & (\sqrt{2})^{\lg n} & n^2 & n! & (\lceil \lg n \rceil)! \\
\left(\frac{3}{2}\right)^n & n^3 & \lg^2 n & \lg(n!) & 2^{2^n} & n^{1/\lg n} \\
\lg\lg n & \lg^* n & n2^n & n^{\lg\lg n} & \log n & 1 \\
2^{\lg n} & (\lg n)^{\lg n} & e^n & 4^{\lg n} & (n+1)! & \sqrt{\lg n} \\
\lg^*(\lg n) & 2^{\sqrt{2\lg n}} & n & 2^n & n\lg n & 2^{2^n+1}
\end{array}$$

Answer:

First of all, we should simplify some of the expressions appearing, for instance:

$$(\sqrt{2})^{\lg n} = 2^{\lg n^{1/2}} = n^{1/2} \tag{1}$$

and:

$$n^{1/\lg n} = (2^{\lg n})^{1/\lg n} = 2 \tag{2}$$

$$2^{\lg n} = n \tag{3}$$

Now, we know that the slowest are the constants, so we have 1 and 2, then we will have to consider logarithms and iterative logarithms, and we can see that, because of the definition of $\lg^*$ we have that:

$$\lg^* \lg n \sim \lg^* n - 1 \sim \lg^* n \tag{4}$$

because we have already iterated once by using $\lg n$, we need one less to arrive to $n \leq 1$.

Then we have also that if $\lg^* n = k$, then we get that:

$$2^{2^{\cdot^{\cdot^{2}}}} \leq n \quad \text{Exponentiation k - 1 times} \tag{5}$$

and then we take $\lg n$ at both sides:

$$k \leq 2^{2^{\cdot^{\cdot^{2}}}} \leq \lg n \quad \text{Exponentiation k - 2 times} \tag{6}$$

and therefore we have that the logarithm grows faster than its iterative version.

Then we can see that:

$$1,\, 2 \ll \lg\lg^* n \ll \lg^* n,\, \lg^* \lg n \ll \lg\lg n \tag{7}$$

because $\lg\lg^* x$ is the composition $f(g(n))$ where $f(n) = \lg n$ and $g(n) = \lg^* n$ whereas $\lg^* n$ is the composition when $f(n) = n$.

Now we should consider the scaling of the logarithms:

$$\lg\lg n \ll \sqrt{\lg n} \ll \lg n \ll \lg^2 n \tag{8}$$

After this we start to have square roots and linear polynomials:

$$\lg^2 n \ll n^{1/2} \ll n \tag{9}$$

then we can go with the n times the logarithm:

$$n \ll n\lg n,\, \lg(n!) \tag{10}$$

The proof that $\lg n! \sim n \lg n$ is done using Stirling's formula and has been done already in class.

Then we can easily see that:

$$n \lg n \ll n^2 \ll n^3 \tag{11}$$

and we can continue with the exponentials, for which we have the following order relation (you can prove this by taking the limit of the quotient):

$$n^3 \ll (3/2)^n \ll 2^n \ll n2^n \ll e^n \tag{12}$$

after the exponentials, we have the factorial regime, for which we have, using Stirling's formula and quotients that:

$$e^n \ll n! \ll (n+1)! \tag{13}$$

and after this we have the super exponentials, for which is easy to see that:

$$(n+1)! \ll 2^{2^n} \ll 2^{2^{n+1}} \tag{14}$$

therefore we are left to clasify the following which are not that easy at first glance:

$$2^{\lg^* n},\ n^{\lg \lg n},\ (\lg n)^{(\lg n)},\ (\lceil \lg n \rceil)!,\ 2^{\sqrt{2 \lg n}} \tag{15}$$

We can easily classify $2^{\lg^* n}$ using again the tower exponentiation trick and see that it's:

$$\lg^* n \ll 2^{\lg^* n} \ll \lg \lg n \tag{16}$$

With regards to $n^{\lg \lg n}$ we can take logs and see:

$$k \lg n \ll \lg n \lg \lg n \ll \lg^2 n \tag{17}$$

and then exponentiating again:

$$n^k \ll n^{\lg \lg n} \ll 2^{\lg^2 n} \tag{18}$$

So it's faster than all of the polynomials, however, it is slower than the smallest of the rest of the exponentials.

We see also that $(\lg n)^{\lg n}$ has the same growth due to taking logarithms.

Regarding $(\lceil \lg n \rceil)!$ we can see using Stirling formula that its position is:

$$n^3 \ll (\lceil \lg n \rceil)! \ll (\lg n)^{\lg n} \tag{19}$$

As for $2^{\sqrt{2 \lg n}}$, simply taking logarithms allow us to put it:

$$\lg^2 n \ll 2^{\sqrt{2 \lg n}} \ll n^{1/2} \tag{20}$$

So finally, we will have:

$$\begin{multline} 1,\ n^{1/\lg n} = 2 \ll \lg \lg^* n \ll \lg^* n,\ \lg^* \lg n \ll 2^{\lg^* n} \ll \lg \lg n \ll \sqrt{\lg n} \ll \\ \lg n \ll \lg^2 n \ll 2^{\sqrt{2 \lg n}} \ll n^{1/2} \ll n \ll n \lg n,\ \lg(n!) \ll n^2 \ll n^3 \ll \\ (\lceil \lg n \rceil)! \ll (\lg n)^{\lg n},\ n^{\lg \lg n} \ll (3/2)^n \ll 2^n \ll n2^n \ll e^n \ll \\ n! \ll (n+1)! \ll 2^{2^n} \ll 2^{2^{n+1}} \end{multline} \tag{21}$$